

TitanZero — Architecture Blueprint

Purpose

This document defines the technical architecture of the TitanZero platform.

It explains how the system is structured within the codebase and how the major engines, workspaces, and interfaces interact.

The goal is to ensure that:

core engines remain stable

extensions integrate safely

AI reasoning remains governed

tenant isolation is preserved

the platform operates seamlessly as both a desktop application and Progressive Web Application (PWA).

Architectural Layers

TitanZero is built as a layered system.

Each layer performs a specific role.

User Interfaces



PWA Application Shell



Workspace Layer



AI Engine Layer



Automation Layer



Signal Bus



Data Core

Above all operational layers sits Titan Zero, the governance intelligence responsible for evaluating high-risk decisions.

Core Application Structure

TitanZero is built on a Laravel-based application derived from the MagicAI platform.

The core system resides in the primary application directories.

```
app/  
bootstrap/  
config/  
database/  
resources/  
routes/  
public/
```

The Titan engines extend the base framework through a Core Engine Layer.

Core Engine Layer

TitanZero contains several internal engines responsible for signals, automation, AI reasoning, and governance. some engines are below, refer to the engine docs for a full list

```
app/Core/  
  TitanSignal/  
  TitanPulse/  
  TitanDuality/  
  TitanZero/
```

Each engine has a clearly defined role.

Titan Signal

Titan Signal is the event bus of the platform.

Location:

app/Core/TitanSignal

Responsibilities:

record operational events

provide an append-only signal stream

allow engines and extensions to react to events

support automation and lifecycle tracking

Example signals:

job.created

job.completed

invoice.sent

invoice.paid

customer.message.received

campaign.scheduled

Signals form the nervous system of the platform.

Titan Pulse

Titan Pulse is the automation engine.

Location:

app/Core/TitanPulse

Responsibilities:

listen to Titan Signal events

evaluate automation rules

execute automated actions

orchestrate workflow automation

Automation rules follow the pattern:

Trigger → Condition → Action

Titan Pulse functions as the reflex layer of the system.

Titan Duality

Titan Duality is the AI reasoning layer responsible for lifecycle assistants.

Location:

app/Core/TitanDuality

Titan Duality contains two specialised reasoning cores.

LogiCoreAI

CreatiCoreAI

LogiCoreAI

Responsible for operational reasoning and managing operations lifecycle assistants.

CreatiCoreAI

Responsible for creative intelligence and managing marketing lifecycle assistants.

Titan Zero Governance Engine

Titan Zero is the system oversight intelligence.

Location:

app/Core/TitanZero

Responsibilities include:

governance and system oversight

evaluating medium- and high-risk operations

coordinating AI reasoning engines

approving or rejecting significant system actions

strategic planning and system optimisation

Titan Zero normally operates in monitoring mode.

When higher-risk operations occur, the system activates Trinary Bayesian Mode.

Titan Zero Core Architecture

Titan Zero consists of three cooperating reasoning cores.

TitanZeroCore

OmegaCore

OmicronCore

TitanZeroCore

Central coordination and decision synthesis.

OmegaCore

Strategic planning and high-level reasoning.

OmicronCore

Detailed operational analysis and validation.

Trinary Bayesian Mode

When activated:

OmegaCore → strategic analysis

OmicronCore → operational validation

TitanZeroCore → final decision

The system evaluates the decision from multiple perspectives and produces the final approval.

Workspace Layer

TitanZero provides two conceptual workspaces.

Logical Workspace

Operational decision-making environment.

Creative Workspace

Campaign creation and automation design environment.

User Interface Layer

TitanZero provides several user interfaces designed for different operational contexts.

Titan Command

Titan Command is the primary operational command centre.

It is a single-page control panel that provides a real-time overview of the entire business.

Key functions:

operational alerts

dispatch visibility

financial risk monitoring

automation status

system signals

Titan Talk

Titan Talk is the communications interface.

It manages:

customer conversations

internal messaging

AI-assisted replies

conversation threads

communication signals

Titan Talk connects communications directly to operational records such as jobs and customers.

Titan Origin

Titan Origin provides traditional administrative interfaces.

It includes structured screens for managing:

customers

jobs

invoices

system configuration

Titan Origin exists for users who prefer traditional administrative interfaces.

Titan Go

Titan Go is the mobile and field interface of the platform.

Titan Go activates when the system operates in mobile or PWA mode.

It provides simplified operational interfaces optimised for touch interaction and mobile workflows.

Titan Go supports several operational modes designed for different field roles.

Titan Go Operational Modes

Titan Go dynamically switches between operational modes depending on the user's role and context.

Field Mode

Designed for field technicians performing jobs.

Features include:

job details

checklists

evidence capture

job completion actions

customer interaction tools

Command Mode

Mobile version of the Titan Command interface.

Allows owners or supervisors to:

monitor operations

approve actions

respond to alerts

manage business activity from mobile devices.

Dispatch Mode

Used by coordinators managing scheduling and job assignments.

Features include:

job dispatch board

scheduling adjustments

technician assignment

route coordination

QC Mode

Quality control and verification interface.

Features include:

inspection checklists

evidence review

compliance verification

job approval workflows

QC mode ensures that service standards and compliance requirements are maintained.

PWA Application Shell

TitanZero is designed as a Progressive Web Application (PWA).

The PWA shell enables:

installation on mobile devices

standalone application mode

responsive UI behaviour

offline-aware functionality

The PWA architecture allows TitanZero to operate across:

desktop browsers

tablets

smartphones

without requiring separate native applications.

Extension System

TitanZero supports a modular extension architecture.

Extensions are located in:

app/Extensions/

Example extensions include:

MarketingBot
SocialMedia
AiSocialMedia
SocialMediaAgent

Extensions integrate through:

service providers

database tables

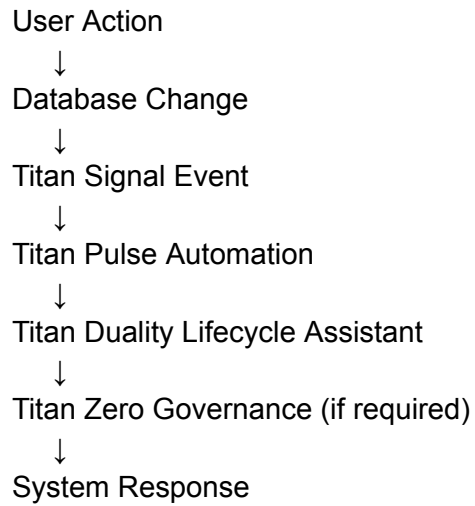
menus

signal listeners

Extensions must follow strict installation and routing rules to maintain system stability and tenant isolation.

Signal Flow

The operational flow of the system follows a signal-driven model.



Summary

TitanZero is structured as a signal-driven architecture composed of specialised engines for signals, automation, AI reasoning, and governance.

Users interact with the system through several interfaces including Titan Command, Titan Talk, Titan Origin, and Titan Go.

The Titan Go interface provides mobile-optimised operational modes such as Field, Command, Dispatch, and QC.

Together these components create a flexible platform capable of operating as both a desktop system and a mobile PWA while maintaining strong automation and AI assistance.